

# Jahangirnagar University

## Institute of Information Technology (IIT)

3rd Year 1st Semester B.Sc. Final Examination 2024

Course Title. Principles of Economics

Course Code. ICT- 3109

Time. 3 Hours

Marks. 60

### Answer to question 1 is mandatory

1. a. What is Economics? Why do you study economics as a student of IIT discipline? Distinguish between Positive and Normative economics with examples. 3
- b. Draw the circular flow diagram of a closed economic system. 4
- c. If a computer firm produces two products (software and hardware) with facing limited investment. Show the efficient, inefficient and unattainable points of the firm by using Production Possibility Frontier (PPF) and why? Explain all the points. 5

### Answer any four from 2-7

2. a. Describe the law of demand for Networking with figure. 3
- b. When the price of computer will increase or decrease and when the demand for computer will increase or decrease, what will happen on the demand curve? Explain graphically. 4
- c. Define market equilibrium. Suppose, A hypothetical market is in equilibrium position- If the market price will increase and will decrease from the equilibrium price, what will happen on the economy-explain (shortage and surplus) with figure. 5
3. a. Distinguish between elastic and inelastic demand with example in each. 4
- b. List all the factors that influence the elasticity of demand. 6
- c. Consider the following demand schedule: 2

| Price | Quantity demand |
|-------|-----------------|
| 50    | 500             |
| 40    | 800             |

Calculate the price elasticity of demand and interpret your result.

4. a. What is aggregate demand curve and aggregate supply curve? 3
- b. Why does the aggregate demand (AD) curve's slope downward? Explain any two reasons. 5
- c. Explain with appropriate figures what happen of price level and output level through aggregate demand (AD) curve : 4
  - i. If investment expenditures increase
  - ii. If net exports decrease
5. a. What is GDP? Differences between nominal GDP and real GDP. 3
- b. What are the components of GDP? Explain. 3
- c. Below are some data on three goods produced in the economy. 6

| Year | Price of rice(/kg) in Tk. | Quantity of rice (kg) | Price of cloth (/yard) in Tk. | Quantity of cloth (/yard) | Price of flour(/kg) in Tk. | Quantity of flour (kg) |
|------|---------------------------|-----------------------|-------------------------------|---------------------------|----------------------------|------------------------|
| 2013 | 12                        | 170                   | 15                            | 350                       | 15                         | 150                    |
| 2014 | 15                        | 200                   | 20                            | 400                       | 22                         | 200                    |
| 2015 | 20                        | 450                   | 25                            | 550                       | 24                         | 350                    |

Compute nominal GDP, real GDP, GDP deflator and economic growth rate for each year, using 2013 is the base year.

6. a. Distinguish between nominal GDP and real GDP. 4

b. Consider the following information: 8

| Year | Price of Rice | Quantity of Rice | Price of Salt | Quantity of Salt |
|------|---------------|------------------|---------------|------------------|
| 2020 | 60            | 100              | 20            | 50               |
| 2021 | 70            | 140              | 25            | 70               |
| 2022 | 80            | 180              | 30            | 100              |

(i) Calculate Nominal GDP and Real GDP (consider 2020 as base year).

(ii) Find the GDP deflator each year.

7. a. What do you mean by short run and long run phenomenon? Explain fixed cost, variable cost and total cost. 4

b. Given the data, compute Total cost (TC), Marginal cost(MC), Average fixed cost(AFC), Average variable cost (AVC), Average total cost (ATC) and show those cost curves graphically. 4

| Combination. | Labor | Output | TFC | TVC |
|--------------|-------|--------|-----|-----|
| A            | 0     | 0      | 25  | 25  |
| B            | 1     | 6      | 25  | 50  |
| C            | 2     | 12     | 25  | 75  |
| D            | 3     | 15     | 25  | 100 |
| E            | 4     | 20     | 25  | 150 |
| F            | 5     | 25     | 25  | 175 |
| G            | 6     | 30     | 25  | 225 |

c. Given the following total revenue (TR) and total cost (TC) functions for a firm, maximize profit ( $\pi$ ) for the firm as follows: (i) Find the average revenue (AR); average cost (AC) functions (ii) Set up the profit function ( $\pi$ ). (iii) Find the maximum profit. 4

$$TR = 20Q - 2Q^2, \quad TC = Q^2 + 4Q$$

# Institute of Information Technology

## Jahangirnagar University

3rd year 1st Semester B.Sc. (Hons.) Final Examination, 2023

Duration: 3 hrs.

Full Marks: 60

Course Code: ICT 3109

Course Title: Principles of Economics

**Do not write anything on the question paper.**

**There are 7 (Seven) questions. Figures in the right margin indicate marks.**

### Part – A: Mandatory to answer

1. (a) What is Economics? Differences between micro and macroeconomics. Why do you study economics as a student of IIT discipline? 3
- (b) Define GDP (Gross Domestic Products) and GNP (Gross National Product). 3
- (c) Define price elasticity of demand. A company used to sell its product at Tk.20 per unit, the market demand was 150 units. When the price was increased to Tk.35 per unit, its demand decreased 110 units. Calculate the price elasticity of demand. 6

### Part – B: Answer any 4 (Four) questions.

2. (a) What is the meaning of demand in economics? What are the determinants of demand? 3
- (b) Describe the law of demand and prove the demand law with figure. 5
- (c) Difference between shift in demand and movement along the demand curve with figure. 4
3. (a) What are the components of GDP (Gross Domestic Product)? Explain. 3
- (b) Draw and briefly explain the circular flow diagram of GDP. Identify the parts of the model that correspond to the flow of goods and services and the flow of taka for each of the following activities: 5
- i) Ami pays a storekeeper tk. 200 for a quart of milk.
- ii) Rimu earns tk.200 per hour working at a fast food restaurant.
- iii) Simu spends tk.1000 to see a drama at the theatre house in Green Road.
- iv) Sharif earns tk.15,000 from his 10 percent ownership of Acme Industrial.
4. (a) What is aggregate demand curve and aggregate supply curve? 3
- (b) Why does the aggregate demand (AD) curve's slope downward? Explain any two reasons. 5
- (c) Explain with appropriate figures what happen of price level and output level through aggregate demand (AD) curve: 4
- i. If investment expenditures increase
- ii. If net exports decrease
5. (a) What are the differences between GDP deflator and Consumer price index (CPI)? 2
- (b) Below are some data on three goods produced in the economy. 8+2

| Year | Price of rice(/kg) in Tk. | Quantity of rice (kg) | Price of cloth (/yard) in Tk. | Quantity of cloth (/yard) | Price of flour(/kg) in Tk. | Quantity of flour (kg) |
|------|---------------------------|-----------------------|-------------------------------|---------------------------|----------------------------|------------------------|
| 2015 | 10                        | 150                   | 12                            | 250                       | 15                         | 250                    |
| 2016 | 15                        | 300                   | 20                            | 500                       | 20                         | 300                    |
| 2017 | 20                        | 450                   | 25                            | 650                       | 22                         | 350                    |

- i) Compute nominal GDP, real GDP, GDP deflator and economic growth rate for each year, using 2015 as the base year.
- ii) Explain the economic meaning of the above results.

- |                                                                                   |   |
|-----------------------------------------------------------------------------------|---|
| 6. (a) What is opportunity cost? Give one example.                                | 2 |
| (b) What are the factors of production? Explain.                                  | 6 |
| (c) Explain production possibilities frontier (PPF) with example.                 | 4 |
| 7. (a) Define inflation. Distinguish between demand-pull and cost-push inflation. | 6 |
| (b) What are the causes of inflation in Bangladesh? Explain.                      | 6 |

# Institute of Information Technology (IIT)

## 3rd Year 1st Semester B.S. (Hons.) Final Examination 2022

Course Title: Principles of Economics

Course Code: ICT-3109

Time. 3 Hours

Marks. 60

### [Answer any five from the following questions]

1. a. A university produces two commodities: research and teaching. The resources the university uses include faculty and staff, libraries, classrooms, labs and so on. The following schedule indicates some points on the university's PPF: (12)

|          | A   | B   | C   | D   | E   | F   | G   |
|----------|-----|-----|-----|-----|-----|-----|-----|
| Research | 900 | 750 | 600 | 450 | 300 | 150 | 0   |
| Teaching | 0   | 20  | 45  | 75  | 110 | 150 | 200 |

- i) Define PPF. Does research production by the university exhibit increasing, constant or decreasing per unit opportunity costs?
  - ii) Define and graph the university's PPF and indicate efficient, inefficient and impossible points.
  - iii) Suppose the university is at point B but would like to alter production to point C. What would be the per-teaching unit opportunity cost of producing the extra teaching units?
  - iv) What will happen to the PPF if the main laboratory partially burns down? Graph the new PPF.
  - v) What will happen to the PPF if all of the campus resources are doubled? Graph the new PPF.
2. a. What are the basic problems of micro economics? Explain. (3)
- b. What is demand? Why does the demand curve downward sloping? Explain with graph. (3)
- c. Given a demand  $Q_d = 100 - 5P$  ; where  $Q_d$  is the quantity demanded and  $P$  is the price. and a supply equation  $Q_s = 80 + 5P$  ; where  $Q_s$  is the quantity supplied and  $P$  is the price, calculate the equilibrium price and quantity and show it in a figure. (6)
3. a. Draw and briefly explain the circular flow diagram. Identify the parts of the model that correspond to the flow of goods and services and the flow of taka for each of the following activities (8)
- i) Sorna pays a store keeper tk.200 for a quart of milk.
  - ii) Ontu earns tk.100 per hour working at a fast food restaurant.
  - iii) Rimu spends tk.500 to see a drama at the theatre house in Green Road.
  - iv) Arif earns tk.10,000 from his 10 percent ownership of Acme Industrial.
- b. Distinguish between shift in demand and movement along the demand curve with figure. (4)
4. a. What is supply? Draw the supply curve for a computer. If (i) technological innovation occurs and (ii) raw materials prices increase in the economy, what will happen in the supply curve? Explain with relevant figures. (6)
- b. Differences between price elasticity of demand and price elasticity of supply? A company used to sell its product at Tk.120 per unit, the market demand was 50,000 units. When price was increased to Tk.190 per unit, its demand decreased 40,000 units. Calculate the price elasticity and the effect of total revenue if demand is met in full at both the prices. (6)
5. a. What do you mean by short run and long run phenomenon? Explain fixed cost, variable cost and total cost. (2)
- b. Differences between perfect competition market and monopoly market with examples. (3)
- c. Given the following total revenue (TR) and total cost (TC) functions for a firm, maximize profit ( $\pi$ ) for the firm as follows: (7)
- (i) Find the average revenue (AR), marginal revenue (MR), average cost (AC) and marginal cost(MC) functions.
  - (ii) Set up the profit function ( $\pi$ ).

(iii) Find the critical value(s) where  $\pi$  is at a relative extremum and test second order condition and calculate the maximum profit.

$$TR = 1400Q - 7.5Q^2, \quad TC = Q^3 - 6Q^2 + 140Q + 750$$

6. a. The consumption bundles of the typical middle-class families of a city include 4 kgs of mangoes, 2 kgs of oranges, 2 kgs of apples, 1 kg of grapes. The market prices of these commodities are as follows: (base year 2001) **(8)**

| Year | price of mangoes | price of oranges | price of apples | prices of grapes |
|------|------------------|------------------|-----------------|------------------|
| 2001 | 25               | 40               | 45              | 70               |
| 2002 | 29               | 45               | 52              | 85               |
| 2003 | 32               | 49               | 60              | 94               |
| 2004 | 35               | 56               | 64              | 100              |
| 2005 | 40               | 65               | 70              | 120              |

Calculate the consumer price index (CPI) and the inflation rate. What are the problems of calculating CPI?

b. What does the GDP deflator and CPI measure? In what respects GDP deflator is different from the CPI. **(4)**

7. a. Define AD (aggregate demand) and AS (aggregate supply) with graph. Why does long run aggregate supply curve vertical? **(8)**

b. Explain with appropriate figures what happen of price level and output level through aggregate demand (AD) curve: **(4)**

i. Bangladesh Govt. undertakes massive investment in physical and social infrastructure development projects.

ii. Net export level increases in the economy

c. Distinguish between demand pull inflation and cost push inflation with graph. **(4)**

# Jahangirnagar University

## 3rd Year 1st Semester B.Sc. (Hons.) Final Examination 2021

COURSE CODE: IT-3109

COURSE TITLE: PRINCIPLES OF ECONOMICS

TOTAL MARKS: 60

TIME: 3 Hours

### ANSWER ANY FIVE (5) QUESTIONS

1. a) Suppose there are six workers in an economy. A worker can produce either five cakes or three shirts per day. 12.0
- i. Draw society's PPF.
  - ii. Why points inside the PPF are inefficient?
  - iii. Why points outside the PPF are unattainable? Explain.
  - iv. Why PPF shifts outward to the right?
  - v. Why PPF is straight line in this case?
2. a) 'Scarcity implies choice; choice implies opportunity cost'-explain. 4.0
- b) What are the resources in the economy? Who are the owners of these resources? What are the rewards of these resources in production? 4.0
- c) Explain the characteristics of 'land' and 'labor' as factors of production. 4.0
3. a) Using supply and demand graphs and assuming competitive markets, show and explain the effect on equilibrium price and quantity of the following: 12.0
- i. Increased prices of milk and sugar on the market for an ice-cream cone.
  - ii. Incidence of bird flu on the market for chicken.
  - iii. A hot weather on the market for ice cream cones.
  - iv. Bumper production on the market for rice.
4. a) State the law of Diminishing Marginal Product. 4.0
- b) Define marginal cost (MC) and average total cost (ATC). Explain the relationship between MC and ATC. 4.0
- c) Why ATC curve is U-shaped? 4.0
5. Suppose in a hypothetical economy, three goods are produced: Mango, Banana and Chicken. The quantity of output and prices of mango, banana and chicken are given in the following Table.
- |         | Output 2014 | Output 2015 | Output 2016 | Price,Tk/<br>Unit 2014 | Price,Tk/<br>Unit 2015 | Price,Tk/<br>Unit 2016 |
|---------|-------------|-------------|-------------|------------------------|------------------------|------------------------|
| Mango   | 150         | 160         | 165         | 105                    | 110                    | 115                    |
| Banana  | 510         | 515         | 530         | 7                      | 8                      | 10                     |
| Chicken | 315         | 320         | 325         | 110                    | 115                    | 125                    |
- a) Define nominal and real GDP. Calculate the nominal GDP for the year 2014, 2015 and 2016. Suppose, 2015 is the base year, and then calculate the real GDP for the year 2014, 2015 and 2016. Define growth rate. Calculate the growth rate of real GDP for the year 2015 and 2016. 6.0
- b) Define GDP deflator. What does it measure? According to the GDP deflator, how much prices have increased for the year 2015 and 2016. 6.0
6. a). Define price elasticity of demand. Briefly explain its determinants? 5.0
- b). On Tuesday, price and quantity demanded are Tk. 7 and 120 units, respectively. Ten days later, price and quantity demanded are Tk. 5 and 150 units, respectively. Calculate the price elasticity of demand and interpret the result. 7.0
7. a) Why is there structural unemployment? Explain. 6.0

b) Suppose the population of Bangladesh is 180 million. Of which 80% are adult. The population not in labor force is 40 million and the number of employed is 90 million. Calculate the labor force participation rate and unemployment rate.

**6.0**